

Internship Report

Data Curation and Scenario Generation used for Investigating the Effectiveness of Novel Machine Learning Algorithms in Passive Acoustic Monitoring (PAM) and Species Diversity Estimation in Ensiferan Community



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CERTIFICATE

This is to certify that Mr. **Arman Singh Jaryal** of the National Institute of Science Education and Research (NISER), Bhubaneswar has successfully completed his summer internship during the period of June–July 2026 on the topic of "**Data Curation and Scenario Generation used for Investigating the Effectiveness of Novel Machine Learning Algorithms in Passive Acoustic Monitoring (PAM) and Species Diversity Estimation in Ensiferan Community**" at Evolutionary Ecology Lab , School of Biological Sciences.



National Institute of Science Education and Research (NISER), Bhubaneswar

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DECLARATION

I hereby declare that I am the sole author of this **Report** in partial fulfillment of the requirements for a **Summer Internship Project** from National Institute of Science Education and Research (NISER), Bhubaneswar.

Signature of the Student

Date:

The project work reported in the report entitled “***Data Curation and Scenario Generation used for Investigating the Effectiveness of Novel Machine Learning Algorithms in Passive Acoustic Monitoring (PAM) and Species Diversity Estimation in Ensiferan Community***” was carried out under my supervision, in the Centre of Ecological Sciences at National Institute of Science Education and Research (NISER), Bhubaneswar .

Signature of the Guide/Supervisor

School:

Date:

ACKNOWLEDGMENTS

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Introduction

In ecology, we study how an organism interacts with its environment and other organisms. Biodiversity estimation serves as the foundation for the analysis of population and community ecology, with traditional techniques often employed to map the abundance, richness, and distribution of individuals within a habitat. The majority of species studies typically rely on active survey techniques, which require physically entering the field to estimate their presence. While camera traps are widely employed for monitoring terrestrial wildlife, they present several limitations for detailed acoustic biodiversity studies. Camera traps are often constrained by limited detection resolution for small or cryptic species, especially those that are not visually distinctive or are morphologically similar. Their ability to detect activity at night or in low-light conditions is primarily restricted to larger endothermic animals due to sensor limitations. Furthermore, camera traps typically fail to record vocalizations, nocturnal or camouflaged individuals, and species with minimal movement, thereby missing critical behavioral and ecological information. But we know that a large number of organisms use **acoustic communication** for vital functions such as attracting mates, signaling danger, and defining territory. This distinct acoustic profile allows the potential differentiation of species merely by their sounds.[1] For these reasons, our approach focuses on acoustic methods, which allow for non-invasive monitoring and identification of a broader range of vocalizing species, providing a richer dataset of behavior and presence in the community.

Study Organism: The Ensiferan Community

Ensiferans (katydids, crickets, and wetas) are the organism of interest in this study. They form a crucial component of many terrestrial ecosystems and utilize species-specific stridulation (sound production) for communication, making bioacoustics a particularly promising survey method for them.

There is a notable lack of published research and reported data on the Biodiversity of Ensiferan community in India. The reason we must study this group is because it is considered a basal community (an older, foundational lineage in the ecosystem) that may provide key insights into evolutionary and ecological processes.[2]

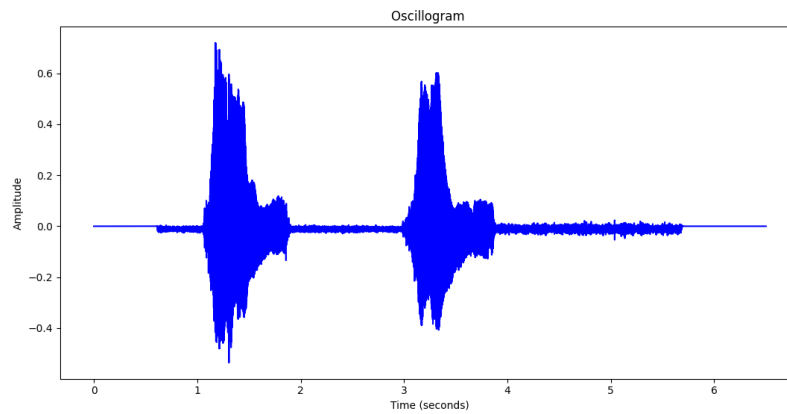
Acoustic Monitoring: Principles and Tools

Acoustic monitoring has emerged as a powerful, non-invasive technique for surveying vocal species. However, traditional bioacoustic monitoring is fraught with challenges. The "gold standard" has historically relied on expert taxonomists to manually identify species by listening to recordings or visually inspecting their **spectrograms** (visual representations of sound). This methodology is constrained by several severe bottlenecks:

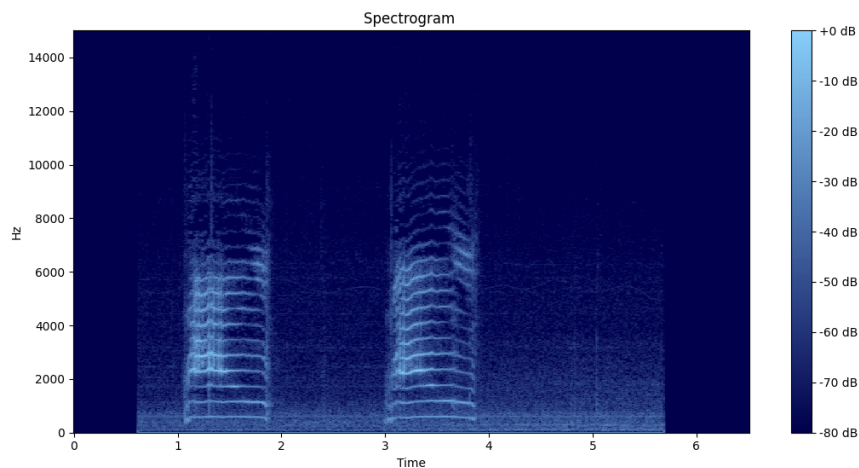
- **Time and Cost:** Manual analysis of thousands of hours of field recordings is prohibitively slow and expensive.
- **Expert Dependency:** The method's success is entirely dependent on the availability of a few, highly-trained experts, creating a "taxonomic impediment."
- **Auditory Limitations:** Many insect species, especially katydids, communicate in the **ultrasonic spectrum**, with frequencies far outside the range of human hearing, making them impossible to identify without specialized equipment and software.[3]

Spectrograms and Oscillograms

A **spectrogram** is a visual representation of the frequency spectrum of a sound signal as it varies over time. The x-axis denotes time (in seconds), the y-axis shows frequency (in Hz), and the color intensity indicates amplitude (in dB). Spectrograms are primary tools for visually identifying species-specific call patterns and frequency structures in biological recordings. An **oscillogram** (or waveform) is the most basic graphical representation of a sound wave, displaying amplitude over time. While less informative for direct species identification than a spectrogram, oscillograms are crucial for detecting large amplitude events, evaluating recording quality, and filtering noise or silent sections in audio data. Figures 1a and 1b below illustrate these visualization techniques, using a recorded example of a cat vocalization captured with a phone microphone. The oscillogram (Fig. 1a) provides a time-domain view, while the spectrogram (Fig. 1b) offers insight into the spectral content and temporal dynamics of the call example.



(a)



(b)

Figure 1: Example of acoustic visualization tools using a cat's vocalization.(a) shows an Oscillogram (waveform) of a cat vocalization, recorded using a phone microphone. The x-axis shows time (seconds) and the y-axis shows amplitude. (b) shows Spectrogram of the same cat vocalization. The x-axis is time (seconds), the y-axis is frequency (Hz), and color intensity represents amplitude (dB). Species-specific call structure is visible as distinct frequency bands.

The Gap in Computational Bioacoustics

To overcome these challenges, **machine learning (ML)** has been successfully applied in other bioacoustic fields. The most prominent examples are in ornithology, where models like:

- **BirdNET**
- **Merlin (from the Cornell Lab of Ornithology)**
- **PERCH (a Google research project)**

use deep neural networks to identify thousands of bird species from their calls with high accuracy. These tools have revolutionized avian monitoring, enabling large-scale, automated data collection.[4][5][6]

Despite the clear need, there are currently no robust, widely adopted, or large-scale ML models for Ensiferan identification. Although software like **Raven Pro** and **Audacity** are invaluable tools for visualizing and annotating spectrograms, they are analysis platforms, not automated identification solutions.[7][8] They do not solve the bottleneck and struggle with the real-world problem of **signal admixture** (overlapping soundscape of multiple species calling simultaneously).

Foundation Model Selection and Core Analysis Principle

The analysis of extensive environmental audio data requires highly robust and generalizable machine learning tools. We have strategically decided to leverage NatureLM-audio as the primary foundation model for core acoustic processing tasks.

This decision is motivated by NatureLM-audio's demonstrated performance, which achieves state-of-the-art results on a wide range of bioacoustics tasks, often exceeding the capabilities of recent models that are highly specialized for their particular training data.[9] Crucially, as a large audio-language model, it exhibits powerful zero-shot generalization to unseen taxa and tasks. This feature eliminates the need for expensive and time-consuming custom model training and fine-tuning for every specific environmental context or species of interest, leading to significant savings in both time and human resource allocation. By adopting this pre-trained foundation model, we ensure a scalable, flexible, and robust analytical pipeline.

Aim

How Effectively can Novel ML Algorithms Estimate Species Diversity in Ensiferan Community?

Key Objectives

- To test run the model to check if the model works by using the provided test examples.
- To find any matching Ensiferan Dataset available from Nature LM's Hugging face Repository and use it to get some insight.
- To acquire data and train the model on it and find it's efficiency.
- To generate data by purifying the data and making it suitable for the model to train on.
- To make complex scenario to simulate the real world challenge of identifying Richness and abundance of the species.

NatureLM-audio: Acoustic Event Detection and Classification

The NatureLM framework processes raw audio data through a sequence of steps designed to isolate and identify relevant biological sounds. This is primarily a two-stage process:

1. **Detection (Event Localization):** The model continuously analyzes the incoming acoustic stream. The absence of an acoustic event of interest is treated as silence or background noise (no signal). When a distinct bioacoustic pattern is encountered that exceeds a learned threshold, the model performs Event Detection. This isolates the sound source (e.g., a bird call, frog vocalization, or marine mammal click) and localizes it to a precise time window within the recording. This effectively distinguishes bioacoustic signal from background noise.
2. **Classification (Event Identification):** Once an event is detected, the model uses its deep knowledge base, derived from its multimodal training, to perform Classification. This involves defining the event by identifying the species, call-type, or other contextual information associated with the localized sound. Classification is achieved through the model's ability to respond to natural language prompts (e.g., "What species is calling in this clip?"), generating a descriptive label (e.g., *Turdus migratorius* or "American Robin song").

Data Acquisition

Acquisition of high-quality audio data is a critical component in this study. I am utilising a curated dataset provided by Souradeep Dutta, a PhD researcher in our lab, which is collected from three dry deciduous field sites Garh Jungle, Illambazar jungle and Ballavpur wildlife Sanctuary in West Bengal, as shown in Figure 2 where the white demarkated zones are the field sites.

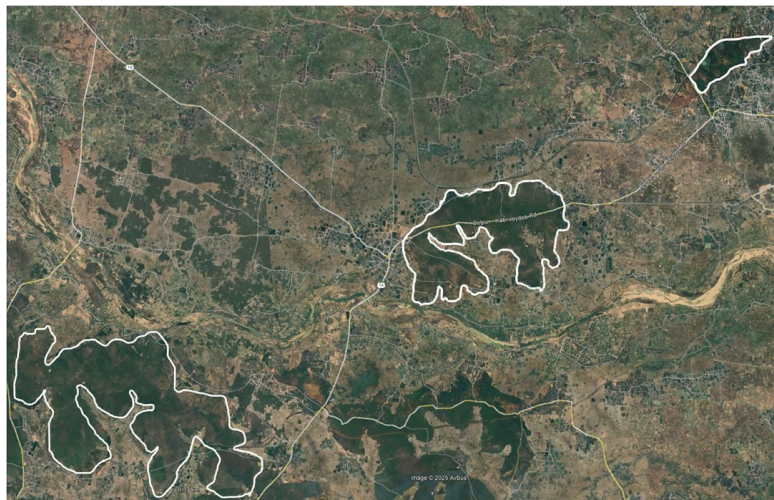


Figure 2: Satellite imagery of the three dry deciduous field sites Garh Jungle, Illambazar jungle and Ballavpur wildlife Sanctuary in West Bengal shown from left to right respectively.

This dataset comprises individual recordings of approximately 30 distinct individuals, each differing in call structure and key acoustic parameters. The majority of these recordings are approximately one minute in duration. For the purposes of model training, we will use 10-second segments extracted from these recordings, aligned with the specific requirements of the machine learning model.

Table 1: Each audio files include around 1 to 2 minute of focal recording of each species

S.No.	Species / OTU Designation	Number of Audio Files
1	<i>Bell cricket</i>	10
2	<i>Euconocephalus</i> sp.	5
3	<i>Gryllodes sigillatus</i>	20
4	<i>Gymnogryllus</i> sp.	9
5	<i>Hexacentrus</i> sp.	5
6	<i>Homeoxipha</i> sp.	12
7	<i>Mecopoda</i> sp.	16
8	<i>Modicogryllus</i> sp.	11
9	OTU 1	4
10	OTU 10 (Type 4)	4
11	OTU 11 (Golden cricket)	11
12	OTU 12	24
13	OTU 13 (Similar to Ballavpur)	24
14	OTU 14 (March special)	4
15	OTU 15	18
16	OTU 16 (Leaf katydid)	18
17	OTU 17	1
18	OTU 24	3
19	OTU 27	7
20	OTU 2 (Flat face)	23
21	OTU 3 (Nemo long)	12
22	OTU 4 (Sand mimic)	3
23	OTU 5	4
24	OTU 7	12
25	OTU 8	7
26	OTU 9 (Type 3)	13
27	<i>Satbrophylla</i> sp.	5
28	Similar to Bell	3
29	Similar to <i>Homeoxipha</i>	6
30	<i>Tarbinskelius</i> sp.	16
31	<i>Teleogryllus</i> sp.	14
32	<i>Xenogryllus</i> sp.	1

Table 2: 10 seconds Recording split per species from 1-2 minute focal recording

S.No.	Species / OTU Designation	Number of Audio Files
1	<i>Bell cricket</i>	68
2	<i>Euconocephalus</i> sp.	49
3	<i>Gryllodes sigillatus</i>	56
4	<i>Gymnogryllus</i> sp.	22
5	<i>Hexacentrus</i> sp.	47
6	<i>Homeoxipha</i> sp.	124
7	<i>Mecopoda</i> sp.	164
8	<i>Modicogryllus</i> sp.	122
9	OTU 1	51
10	OTU 10 (Type 4)	30
11	OTU 11 (Golden cricket)	71
12	OTU 12	256
13	OTU 13 (Similar to Ballavpur)	168
14	OTU 14 (March special)	22
15	OTU 15	188
16	OTU 16 (Leaf katydid)	141
17	OTU 27	71
18	OTU 2 (Flat face)	188
19	OTU 3 (Nemo long)	113
20	OTU 4 (Sand mimic)	19
21	OTU 5	37
22	OTU 7	113
23	OTU 8	101
24	OTU 9 (Type 3)	65
25	<i>Sathrophyllia</i> sp.	45
26	Similar to Bell	2
27	Similar to <i>Homeoxipha</i>	44
28	<i>Tarbinskelius</i> sp.	153
29	<i>Teleogryllus</i> sp.	93
30	<i>Xenogryllus</i> sp.	10

Experiments

Before diving into our full dataset, we conducted a preliminary trial run using audio from species that were already integrated into the model to establish a baseline for performance and we got to know that the model is indeed working. After that, I got the focal recordings for 32 specific species, as detailed in Table 1; these were gathered by actively approaching individual animals in the field by the lab PhDs Souradeep Dutta and Dilshad K to ensure the highest possible signal clarity. While these focal recordings were much cleaner than Passive acoustic Recording, though they still contained inevitable traces of environmental noise. We used several 10-second segments

from these recordings to test whether the model's audio encoder could successfully distinguish between different individuals and species based on these short acoustic bursts. However, we recognized that relying on such clean data wouldn't give us a realistic picture of the model's ability to handle true acoustic diversity in the wild. Real-world passive acoustic monitoring is often messy and full of overlapping sounds, so we couldn't just assume success in the lab would translate to success in the field. To address this, we designed a series of artificial scenarios where we systematically added layers of complexity and noise to the data. This was done to simulate the difficult conditions of a noisy environment, ensuring a much more robust transition from controlled laboratory classification to the unpredictable reality of long-term ecological monitoring. These are the 6 different acoustic scenarios we have created:

- 1. Clean Data
- 2a. Temporal Phase Overlap
- 2b. Pair species without SNR
- 2c. Chorus Test
- 3. Pair species SNR Test
- 4. Multiple Species

Scenario 1: Clean Data

Scenario 1 is the clean baseline where you have the unaltered audio first for 30 species, later denoised removed the species from the list with very noisy data which reduced it to 25 and soon to 21 species. We use it to measure the model's core species recognition ability before introducing harder acoustic conditions.

Scenario 2a: Temporal Phase Overlap

Scenario 2a mixes two units of the same species across overlap levels from 0% to 100% (temporal overlap), without adding noise. We use it to test the accuracy of the identification with varying timing overlap two same-species callers are active together.

Scenario 2b: Pair Species Without SNR / Same-Species Pair with Noise Sweep

In the implemented benchmark, Scenario 2b is two units of the same species with overlap plus SNR stress (20, 10, 5, 0 dB). We use it to quantify robustness to background noise while keeping species identity fixed, so any drop is mostly acoustic (not cross-species confusion).

Scenario 2c: Chorus Test

Scenario 2c simulates chorus behavior by stacking multiple callers from the same species (higher unit density), optionally under noise/overlap variation. We use it because real field recordings

often contain many simultaneous callers of the same species, which is especially hard for counting.

Scenario 3: Pair Species SNR Test

Scenario 3 mixes one unit from species A and one unit from species B, then varies overlap and SNR. We use it to test cross-species separation under interference, which is closer to natural mixed soundscapes than single-species clips.

Scenario 4: Multiple Species

Scenario 4 is the full-complexity stress test: multi-species, multi-unit mixtures (typically 2 to 10 species with multiple units per species), with overlap and noise variation. We use it to approximate real ecological recordings where many taxa vocalize simultaneously.

Initial Results

When identifying 30 distinct species from their audio recordings, the model achieved a species classification accuracy of 78.56% (5,445/6,931 correct) and a unit count accuracy of 88.00% (6,099/6,931 correct). We did a t-SNE run for clean recording to get a visual representation of how model is distinguishing each species based on their clean audio features. A t-Distributed Stochastic Neighbor Embedding (t-SNE) is a machine learning algorithm used for dimensionality reduction, specifically designed to visualize complex, high-dimensional data. Following the exclusion of noise and species (from 30 to 25 species) with data from the available pool, the model achieved a total accuracy of 98.32%, an improved performance that is clearly reflected by the dense, well-isolated clusters in the updated t-SNE visualization as mentioned in the Figure 4.

Audio embeddings are massive, complex arrays of numbers that are impossible to interpret just by looking at the raw data. t-SNE is used here as a diagnostic and validation tool. By squashing those high-dimensional acoustic signatures into a 2D scatter plot, you can visually verify if your model is actually learning to tell the crickets apart. If the algorithm successfully maps the calls of the same species into tight, isolated groups, it proves that the acoustic features you are using are highly discriminative and your classification pipeline is working effectively.

Table 3: The 25 cricket species/OTUs selected for filtered dataset after removing noisy recordings from the initial 30 species pool.

S. No.	Species / OTU Name
1	<i>Bell Cricket</i>
2	<i>Euconocephalus</i> sp.
3	<i>Gryllodes sigillatus</i>
4	<i>Gymnogryllus</i> sp.
5	<i>Hexacentrus</i> sp.
6	<i>Homeoxipha</i> sp.
7	<i>Mecopoda</i> sp.
8	<i>Modicogryllus</i> sp.
9	OTU 1
10	OTU 2 (Flat face)
11	OTU 3 (Nemo long)
12	OTU 5
13	OTU 7
14	OTU 8
15	OTU 9 (Type 3)
16	OTU 11 (Golden cricket)
17	OTU 12
18	OTU 13 (Similar to Ballavpur)
19	OTU 15
20	OTU 16 (Leaf katydid)
21	OTU 27
22	<i>Satherophyllia</i> sp.
23	<i>Tarbinskiellus</i> sp.
24	<i>Teleogryllus</i> sp.
25	Similar to <i>Homeoxipha</i>

Scenario 1: Clean Data

For the 25-species dataset (1,049 clean recordings), Scenario 1 evaluated 1,049 samples and achieved 93.52% species accuracy (981/1,049) with 0.00% count accuracy in that run. For the 21-species dataset (1,651 recordings), Scenario 1 evaluated 1,651 samples and achieved 79.35% species accuracy (1,310/1,651) and 99.88% count accuracy (1,649/1,651). This baseline is important because it tells us whether errors are due to intrinsic class confusion or due to overlap/noise stress.

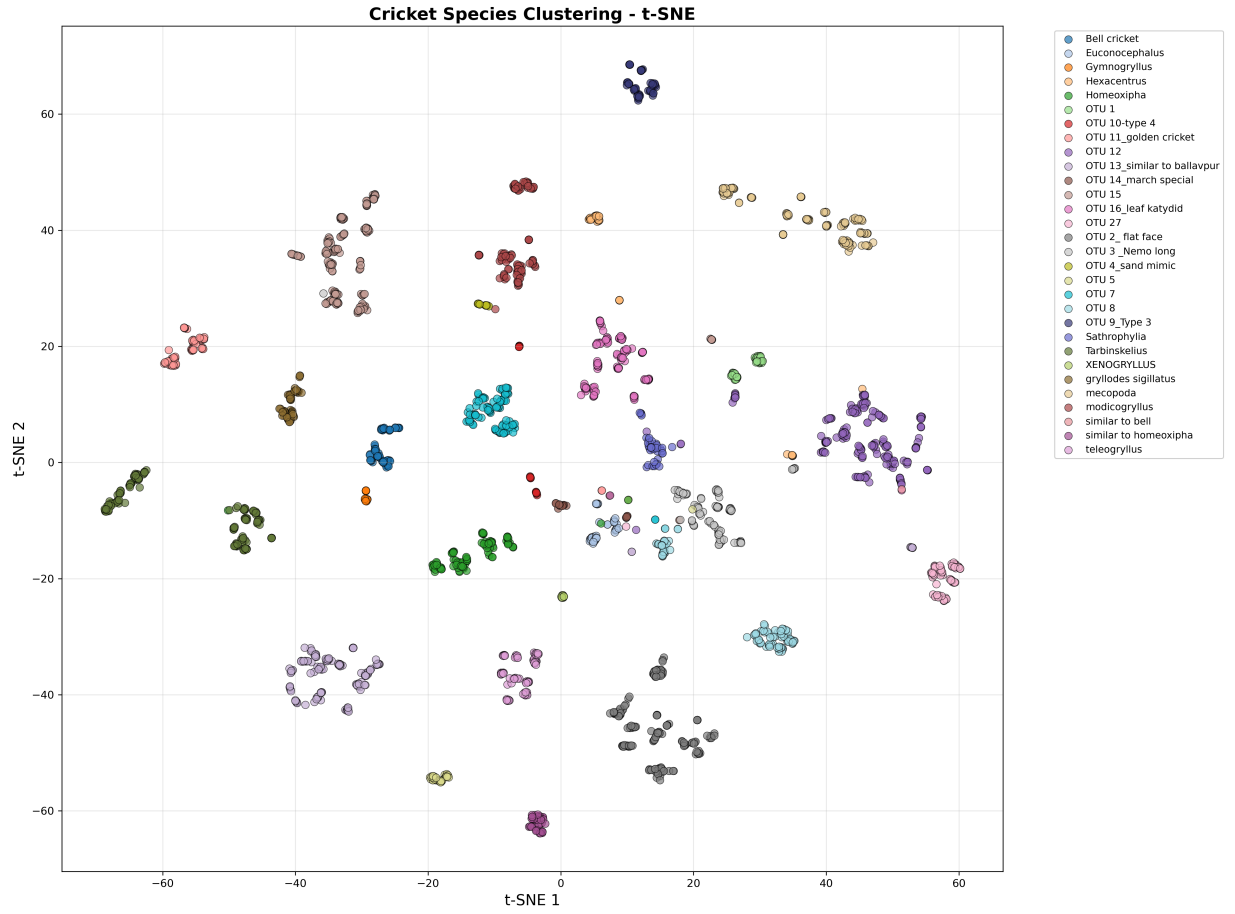


Figure 3: t-SNE visualization of optimized acoustic embeddings for 30 cricket species and OTUs.

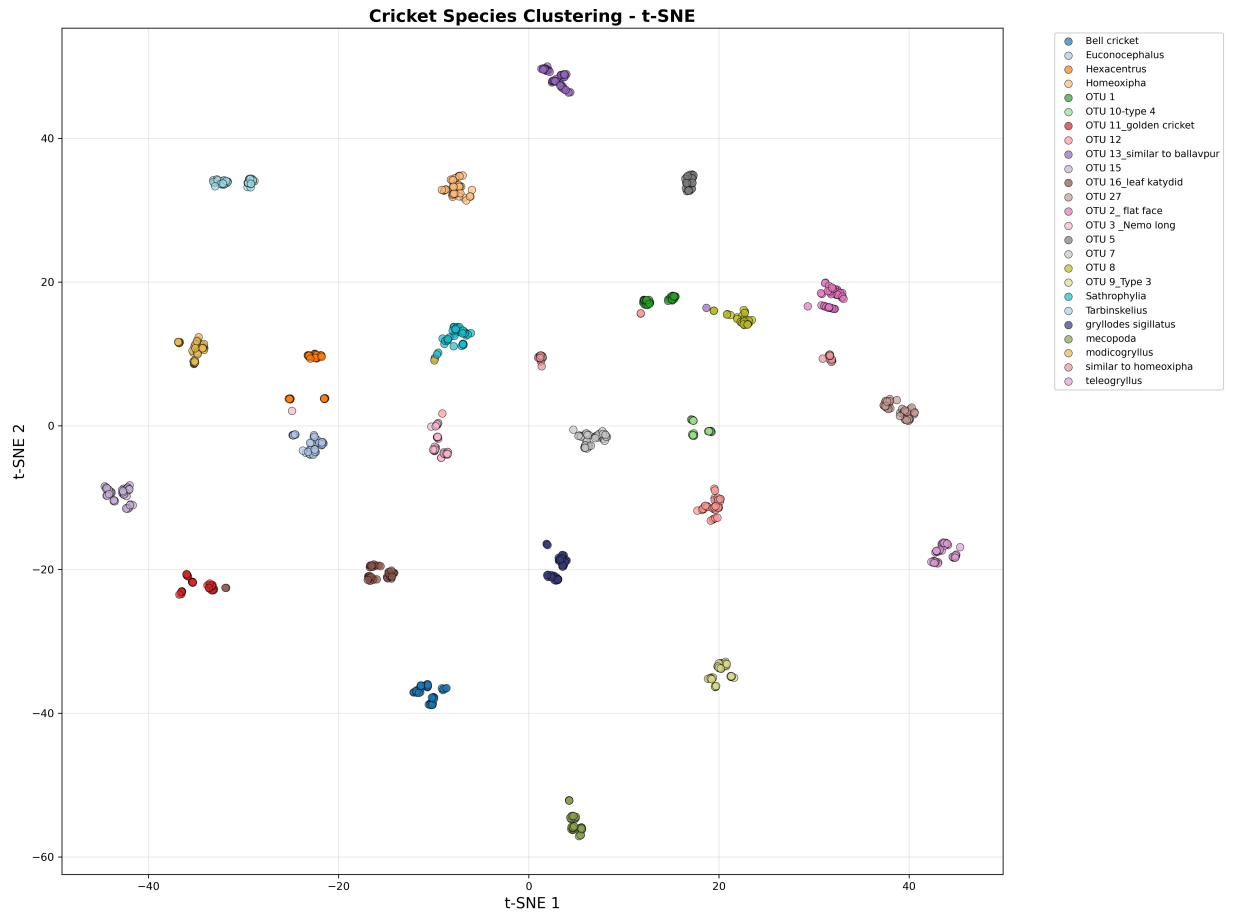


Figure 4: t-SNE visualization of acoustic features for 25 cricket species and OTUs.

Table 4: Model Performance across Scenarios

Scenario	Samples	Species Acc (%)	Count Acc (%)
For 25 Species			
1	1049	97.521449	99.23736892
2a	50500	99.9980198	99.84950495
2b	101000	99.98910891	99.24653465
2c	202000	99.98762376	93.68514851
3	151500	99.91023102	99.73927393
4	1136250	99.86314631	99.38464246
For 21 Species			
1	1651	79.345851	99.8788613
2a	42420	100	100
2b	84840	100	99.88920321
2c	169680	99.9392975	96.90299387
3	106050	99.91023102	99.79255068
4	1136250	99.85020902	99.47837184

Scenario 2a: Temporal Phase Overlap

In the 25-species evaluation, Scenario 2a had 50,500 samples and reached 99.99% species accuracy and 99.39% count accuracy. In the 21-species evaluation, Scenario 2a had 42,420 samples and reached 100.00% species accuracy and 100.00% count accuracy. This shows the model is extremely robust to phase/overlap effects by themselves.

Scenario 2b: Pair Species Without SNR / Same-Species Pair with Noise Sweep

For 25 species, Scenario 2b evaluated 101,000 samples with 99.99% species accuracy and 98.60% count accuracy. For 21 species, Scenario 2b evaluated 84,840 samples with 100.00% species accuracy and 99.89% count accuracy. Performance stays very high, with only small degradation in count prediction at harder SNR.

Scenario 2c: Chorus Test

In the 25-species setting, Scenario 2c had 202,000 samples, with 99.97% species accuracy and 90.07% count accuracy. In the 21-species setting, Scenario 2c had 169,680 samples, with 99.94% species accuracy and 96.90% count accuracy. Species detection remains excellent, but this scenario exposes count-estimation difficulty under dense chorus conditions.

Scenario 3: Pair Species SNR Test

In the 25-species evaluation, Scenario 3 used 151,500 samples and achieved 99.66% species accuracy and 99.27% count accuracy. In the 21-species evaluation, Scenario 3 used 106,050 samples and achieved 99.93% species accuracy and 99.79% count accuracy. These results indicate strong multi-label separation even with overlap and noise.

Scenario 4: Multiple Species

In both evaluations, Scenario 4 had 1,136,250 samples. For 25 species, it achieved 99.80% species accuracy and 98.64% count accuracy; for 21 species, it achieved 99.85% species accuracy and 99.48% count accuracy. Strong performance here shows the model scales from controlled settings to highly complex mixtures with limited degradation.

Critical Reassessment in Noise Modeling, and Dataset Architecture

While the initial benchmark evaluations in Table 4 yielded exceptionally high classification accuracy across artificial scenarios, a critical re-evaluation of the data generation methodology exposed a fundamental conceptual confound, prompting a complete redesign of our noise degradation and dataset architecture.

Data Processing and Focal Dataset Curation

Training our model on long-form field recordings presented a major challenge, primarily because raw audio is rarely consistent and the abundance of the target signal can shift unpredictably throughout a single file. For instance, a typical one-minute recording might contain long stretches of silence where the animal simply stops calling, and it almost always concludes with the researcher’s voice announcing metadata like the temperature, time, and location—all of which would confuse the model if left in. To fix this, we used a custom Python script to curate the data into manageable 10-second snippets, specifically designed to capture the unique acoustic features of each species. We didn’t just rely on the script, though; we used Raven Pro and Audacity to manually inspect every single clip, looking at the spectrograms and listening to the audio to ensure we were keeping only the high-quality signals and discarding anything with excessive noise or data of undesired species.

This rigorous processing and cleaning workflow proceeded in stages, narrowing the species pool to ensure high signal clarity and sufficient recording volume per species.

Manual Data Cleaning of Focal Recordings

A dedicated phase of data curation focused on a core dataset of 152 manually cleaned focal recordings across 16 cricket species. Individual recording durations ranged from approximately 15 seconds

to 4 minutes, with a mean duration of 1 minute 30 seconds. Table 5 details the distribution of original focal recordings and final cleaned recordings across these 16 species.

Table 5: Distribution of original and cleaned focal recordings across 16 cricket species.

S.No.	Species / OTU Name	Original Recordings	Cleaned Recordings
1	<i>Bell cricket</i>	10	7
2	<i>Euconocephalus</i> sp.	5	4
3	<i>Gryllodes sigillatus</i>	20	14
4	<i>Gymnogryllus</i> sp.	9	8
5	<i>Mecopoda</i> sp.	16	9
6	<i>Modicogryllus</i> sp.	11	10
7	OTU 1	4	4
8	OTU 2	22	15
9	OTU 3	12	6
10	OTU 15	18	16
11	OTU 27	7	6
12	OTU 9	13	6
13	<i>Tarbinskelius</i> sp.	16	16
14	<i>Teleogryllus</i> sp.	14	9
15	<i>Homeoxipha</i> sp.	12	6
16	OTU 13	24	16
Total	16 Species	213	152

Cleaning Methodology and Rationale

Cleaning was performed manually in Audacity, working directly on the spectrogram view rather than the waveform view, using parametric equalization (EQ) to attenuate frequency bands dominated by non-target sound sources (background species, wind, anthropogenic noise) while preserving the target species' call band.

A deliberate methodological constraint was imposed throughout: **cleaning was performed conservatively**. The temptation in manual denoising is to be aggressive—pushing the noise floor down as far as possible for a visually and audibly "cleaner" result. This was explicitly avoided because aggressive EQ cleaning artificially inflates the recording's true Signal-to-Noise Ratio (SNR), corrupting downstream quantitative characterization and misrepresenting natural field conditions. The goal was to remove clearly identifiable contaminating sources while leaving the natural relationship between the target call and its residual ambient background intact.

Unsupervised Signal-to-Noise Ratio (SNR) Quantification

To quantitatively evaluate recording quality without human bias, we developed an unsupervised algorithm to compute the SNR of digital recordings directly, without requiring ground-truth signal/noise labels.

1. Adaptive Frequency Band Discovery and Carrier Centering

Cricket calls are produced by stridulation, generating a pulsatile signal with strong temporal energy variance. The algorithm computes the variance of energy across frequency bins over time to discover the primary call band, distinguishing it from low-variance steady-state noise (e.g., wind). Within this band, the algorithm locates the carrier frequency (the peak energy bin) and recenters the analysis band symmetrically around it.

2. Frame-wise Otsu Thresholding and Dynamic Bounds

The audio is divided into 10 ms frames with an 80% overlap (2 ms hop size). Otsu's thresholding method is applied to the frame energy series to find the optimal contrast point between signal and noise candidates. To adapt to drifting background energy over time, a locally-adaptive dynamic threshold is defined as:

$$\text{threshold}(t) = \text{clip}\left(\text{local_mean}(t) \times \text{THRESHOLD_FACTOR}, \text{global_floor}, \text{global_ceiling}\right) \quad (1)$$

where $\text{local_mean}(t)$ is a rolling average computed over a 0.05 s window.

Empirical testing revealed two crucial failure modes in unbounded thresholding, requiring global bounds:

- **Silence-Dominated Recordings (Floor Need):** In clips with long silent stretches (e.g., OTU 1), $\text{local_mean}(t) \rightarrow 0$, causing the threshold to collapse and misclassify background noise as call signal. A global floor was introduced: $\text{global_floor} = P_{30}(\text{rms}) \times \text{FLOOR_MULT}$. Setting $\text{FLOOR_MULT} = 4.5$ for OTU 1 restored accurate segmentation and a clean SNR of 30.78 dB.
- **High-Amplitude Spiky Recordings (Ceiling Need):** In recordings with loud chirps intermixed with quieter calls (e.g., OTU 13), the threshold rose excessively during loud bursts and swallowed quieter calls into noise. A global ceiling was introduced: $\text{global_ceiling} = P_{90}(\text{rms}) \times \text{CEIL_MULT}$ (default 0.40), preventing the threshold from exceeding 40% of peak energy.

3. Time-Domain SNR Computation

Signal samples s and noise samples n are segmented based on the dynamic threshold. Signal power $P_{\text{signal}} = \text{mean}(s^2)$ and noise power $P_{\text{noise}} = \text{mean}(n^2)$ are computed. Because signal frames superpose call energy and ambient noise, the noise power is subtracted to isolate pure cricket call power:

$$P_{\text{cricket}} = \max(P_{\text{signal}} - P_{\text{noise}}, \varepsilon) \quad (2)$$

$$\text{SNR} = 10 \cdot \log_{10} \left(\frac{P_{\text{cricket}}}{P_{\text{noise}}} \right), \quad \frac{A_c}{A_n} = \sqrt{\frac{P_{\text{cricket}}}{P_{\text{noise}}}} \quad (3)$$

Values above 20 dB are interpreted as Excellent, 10-20 dB as Good, 3-10 dB as Poor, and 0 dB as Unusable.

4. Spectral Fallback for Continuously Trilling Species

For continuously trilling species (*Tarbinskelius*, OTU 3, *Euconocephalus*), time-domain silent gaps do not exist. The algorithm automatically switches to a spectral fallback mode, computing power across short-time Fourier transform (STFT) bins:

$$P_{\text{in_band}} = \text{mean}(\text{STFT}[:, \text{in_band_mask}]), \quad P_{\text{out_of_band}} = \text{mean}(\text{STFT}[:, \text{out_band_mask}]) \quad (4)$$

A ± 500 Hz guard zone around the call band edges is excluded from $P_{\text{out_of_band}}$ to prevent spectral leakage from contaminating the noise estimate. SNR is then computed as $10 \log_{10}(P_{\text{cricket}}/P_{\text{out_of_band}})$.

Initial Noise Ladder Degradation

In earlier iterations, additive noise was applied directly to baseline recordings pre-processed with software such as Raven Pro. However, many of these audio files contained interspecies vocalizations and silent or out-of-band temporal segments. To isolate the focal signals, interspecies calls were manually removed, and each recording was trimmed to the specific spectral and temporal bounds of the target species' vocalizations.

To systematically evaluate model robustness against noise, an 11-tier Gaussian white noise ladder was applied to these cleaned focal recordings. For a given recording with a measured baseline signal-to-noise ratio ($\text{SNR}_{\text{baseline}}$), the 11 target SNR tiers were defined as follows:

$$\text{target SNR}[i] = \text{SNR}_{\text{baseline}} \times \frac{11 - i}{11}, \quad i = 1, 2, \dots, 11 \quad (5)$$

Target noise power $P_{\text{noise_target}} = P_{\text{cricket}}/10^{(\text{target}/10)}$ was solved, and Gaussian white noise $N(0, \sigma_N^2)$ with variance $\sigma_N^2 = P_{\text{noise_target}} - P_{\text{noise_baseline}}$ was added to the audio waveform. Adding equal absolute noise energy to both signal and noise frames mathematically reduces net SNR.

The Noise-Character Confound: Why the Earlier Approach Was Wrong

Upon inspecting the benchmark results, we realized that Gaussian white noise injection suffers from a critical flaw: **real environmental background noise is structured and non-random**. Ambient noise carries a distinct "character" or acoustic fingerprint reflecting the specific field location, habitat type, weather conditions, and background biophony.

Because focal recordings for a given species were collected from specific field locations, all recordings of that species inherently shared a common background noise fingerprint. Adding spectrally flat Gaussian white noise lowered the SNR but **failed to decouple or diversify the underlying location-specific background noise**. As a result, the deep neural network learned a shortcut: it memorized the background noise signature associated with the location of each species, rather than learning the intrinsic call structure of the insect itself. This shortcut learning explains the artificially sky-high accuracies ($\sim 99.9\%$) observed across complex artificial scenarios in Table 4.

The New Architecture

To eliminate shortcut learning, we redesigned the dataset generation strategy to prioritize **compositional diversity** and multi-species mixing, ensuring the model encounters every species against diverse, overlapping call backgrounds and varying individual densities.

1. Storage Bottleneck and the Cyclic Shift Solution

Exhaustively enumerating all combinations of species, unit counts, overlap percentages, and noise tiers for 16 species yields 11.85 million files (≈ 11.1 TB), far exceeding our 1 TB hardware budget. Randomly discarding 90% of combinations was rejected because random sampling introduces severe selection bias across species pairs.

To achieve strict species balance within a 1 TB budget, we designed a **Deterministic Cyclic Shift (Modular) Selection Algorithm**:

1. **Seed Blocks:** Define K seed blocks, each containing an array of r species indices ($r \in \{3, 5, 10\}$).
2. **Modular Rotation:** Each seed block is shifted by $+1 \pmod{16}$ across 16 successive cycles, generating $K \times 16$ unique combinations.
3. **Guaranteed Balance:** Because every species index appears exactly once in every rotation cycle, the total exposure frequency of all 16 species across the dataset is provably identical.

2. Refined Generation Scenarios and Corrected Dataset Ledger

The dataset is structured into four scenarios of increasing acoustic complexity:

- **Scenario 1 (Clean Baseline):** $16 \text{ species} \times 500 \text{ files} = 8,000 \text{ base files}$. Corrected multiplier is $\times 14$ (1 raw + 1 clean + 12 noise tiers), yielding 112,000 files (102.4 GB).
- **Scenario 2 (Intra-Species Variance):** $16 \text{ species} \times 5 \text{ unit counts} \times 5 \text{ overlaps} \times 50 \text{ files} = 20,000 \text{ base files}$ ($\times 13 = 260,000 \text{ files} / 237.81 \text{ GB}$).
- **Scenario 3 (Inter-Species Pairs):** $C(16, 2) = 120 \text{ pairs} \times 5 \text{ overlaps} \times 34 \text{ files} = 20,400 \text{ base files}$ ($\times 13 = 265,200 \text{ files} / 242.57 \text{ GB}$).
- **Scenario 4 (Complex Multi-Species Groupings):** Divided into 4 density sub-tiers (2, 3, 5, and 10 species). Tier A uses combinatorial pairs ($\times 7$), while Tiers B, C, and D use Cyclic Shift selection ($K = 26, 52, 52$ respectively), yielding 650,520 files (595.02 GB).

Table 6 details the final, corrected macro-level dataset ledger totaling 1,287,720 files ($\approx 1.18 \text{ TB}$).

Table 6: Corrected macro-level dataset ledger for the 16-species scenario architecture.

Scenario / Tier	Grouping Method	Multiplier	Clean Files	Total Files	Storage
Scenario 1	Direct slice	$\times 14$ (corrected)	8,000	112,000	$\approx 102.40 \text{ GB}$
Scenario 2	Parametric loops	$\times 13$	20,000	260,000	$\approx 237.81 \text{ GB}$
Scenario 3	Combinatorial $C(16, 2)$	$\times 13$	20,400	265,200	$\approx 242.57 \text{ GB}$
Scenario 4, Tier A	Combinatorial $\times 7$	$\times 13$	12,600	163,800	$\approx 149.82 \text{ GB}$
Scenario 4, Tier B	Cyclic Shift ($K = 26$) $\times 2$	$\times 13$	12,480	162,240	$\approx 148.40 \text{ GB}$
Scenario 4, Tier C	Cyclic Shift ($K = 52$) $\times 1$	$\times 13$	12,480	162,240	$\approx 148.40 \text{ GB}$
Scenario 4, Tier D	Cyclic Shift ($K = 52$) $\times 1$	$\times 13$	12,480	162,240	$\approx 148.40 \text{ GB}$
Grand Total	Symmetric Matrix	N/A	98,440	1,287,720	$\approx 1.18 \text{ TB}$

3. File Typologies and 7-Stage Pipeline Orchestration

Four explicit file typologies are generated across the pipeline: Cleaned, Raw, Noised, and **Submerged** (representing audio where extreme noise completely obscures the target call, explicitly training the model to report "no detection" under severe interference).

The master pipeline orchestrates processing across seven sequential stages:

1. **Stage 1 (Cleaning):** Automated call detection and short-time Fourier transform (STFT) masking (10 ms window), logging call timestamps to `Mask_Lib.csv`.
2. **Stage 2 (Standardization):** Resampling all audio to a uniform 48,000 Hz using a heavy-attenuation Kaiser-window filter.
3. **Stage 3 (Segmentation):** Chopping audio into 10-second clips with staggered start offsets (e.g., 0 s, 2 s, 3 s).
4. **Stage 4 (Metadata Generation):** Performing train/validation/test splits (mutually exclusive by source file) and executing cyclic shift combinatorics.
5. **Stage 5 (Timing Correction):** Computing lead-in trims so the primary call begins at $t = 0$ and setting overlap offsets for subsequent callers.
6. **Stage 6 (Mixing):** Summing source waveforms according to Stage 4 recipes and applying gain reduction to prevent clipping.
7. **Stage 7 (Stress Testing):** Applying the 12-tier noise ladder (ranging from -111 dBFS at Tier 1 to -1 dBFS at Tier 12) and relabeling fully drowned calls as *Submerged*.

Evaluation Metrics

We report four complementary metrics on the multi-label species detection task. **Micro-F1** aggregates true positives, false positives, and false negatives across all species and samples before computing precision and recall, and is dominated by frequent classes and common scenarios. **Mean Jaccard index** (intersection-over-union) is computed per sample as $|Y \cap \hat{Y}| / |Y \cup \hat{Y}|$, where Y is the true species set and $\hat{Y}$ the predicted set at threshold 0.5, then averaged over all samples; unlike F1 it penalises false positives and false negatives symmetrically within each sample, and we treat it as the *primary single-number metric* since it rewards exactly-correct mixture identification. **Equal-scenario-weighted Macro-F1** computes Macro-F1 (per-class F1, averaged over species) within each scenario, then averages those four scenario-level scores equally, preventing the test-set-heavy Scenario 4 (52 % of samples) from dominating the aggregate. **AUC-ROC (macro)** measures discrimination ability across all sigmoid thresholds, reported per class then macro-averaged; it is threshold-free and complementary to Jaccard. We additionally report **count accuracy**: exact-match rate of the predicted total caller count (1-15) against ground truth, evaluated independently of species-set correctness.

Current Results

Table 7: Test performance (mean $\pm$ std, 3 seeds unless noted). Eq.-Macro-F1 = equal-scenario-weighted Macro-F1.

S.No.	Scenario / Metric	Jaccard	Micro-F1
1	S1 - Single species	0.915 ± 0.001	0.891 ± 0.002
2	S2 - Same-sp. chorus	0.914 ± 0.001	0.883 ± 0.003
3	S3 - Two-species	0.789 ± 0.001	0.846 ± 0.000
4	S4 - Dense mixture	0.625 ± 0.001	0.739 ± 0.001
5	Eq.-Macro-F1	0.855 ± 0.000	
6	AUC-ROC (macro)	0.906 (S4) - 0.997 (S2)	
7	Count accuracy	54.0 ± 0.1%	
8	Inference latency [†]	136.9 ± 16.2 ms/sample, 7.3 samples/s	
[†] Single seed, 240 samples.			

Table reports per-scenario Jaccard and Micro-F1 on the full held-out test set 99,385 samples across the four scenarios; see Table averaged over 3 seeds. Samples where the source call is fully submerged beneath injected noise label *None*, Section carry no evaluable species-identity signal and are dropped from the corpus entirely upstream of evaluation, rather than scored against an empty expected set.

Jaccard is highly reproducible across seeds ($\text{std} \leq 0.001$ in every scenario). S1 (single-species baseline, Jaccard 0.915) and S2 (same-species chorus, Jaccard 0.914) are essentially identical, showing that correctly detecting an intra-species chorus is no harder than detecting an isolated caller. Complexity-driven decline appears once distinct species must be disentangled: S3 (two-species overlap, Jaccard 0.789) and S4 (dense mixture, Jaccard 0.625) fall progressively below S1/S2, consistent with the added difficulty of resolving overlapping, spectrally-competing calls from multiple species rather than simply counting callers of one species. Count accuracy (54.0% exact-match on a nominally 15-way classification task, of which only 10 unit-count values actually occur in the data) lags species-set metrics, reflecting that recovering the precise number of overlapping callers is harder than identifying which species are present.

Robustness to Noise Degradation

To isolate the effect of the 12-tier water-level noise injection from scenario complexity, we evaluated the full 99,385-sample test set against all three seed checkpoints, aggregating by noise

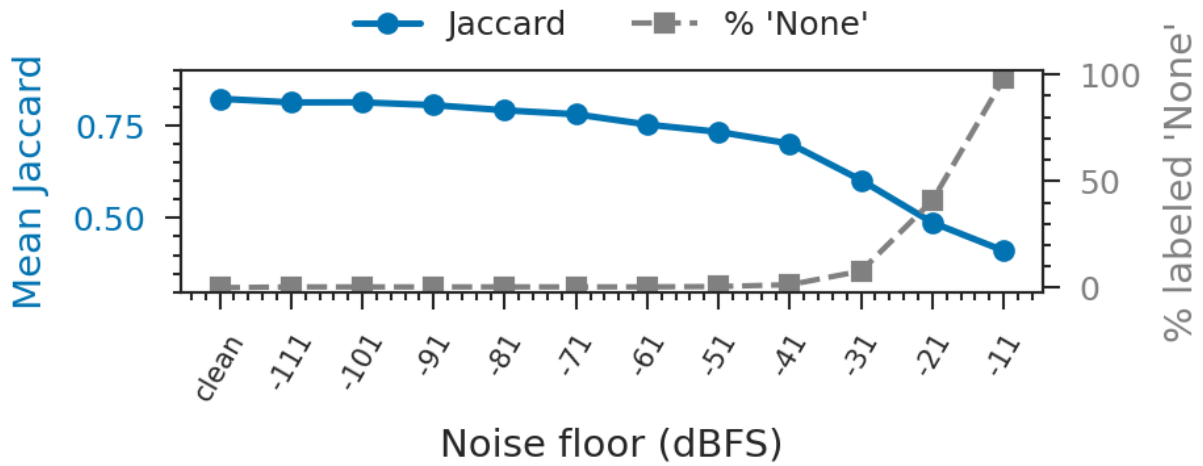


Figure 5: Detection robustness across the noise-injection curriculum (clean plus the 11 tiers with evaluable content; tier 12 is 100% *None* population-wide), full test set (3-seed probability ensemble: sigmoid outputs averaged across seeds, then thresholded once). Left axis: mean Jaccard on samples still labeled with a species. Right axis: population-level rate of samples labeled *None* (signal fully submerged).

tier. plots mean Jaccard on test-set samples still labeled with a species, alongside the *None* rate computed across the full corpus (train+valid+test) at each tier .

Mean Jaccard is 0.823 on clean audio and declines gradually through moderate noise floors, remaining above 0.70 through -41 dBFS (tier 8), then drops more steeply to 0.603 (tier 9, -31 dBFS), 0.488 (tier 10, -21 dBFS), and 0.412 (tier 11, -11 dBFS, only 239 evaluable samples remain). The *None* rate stays under 1.5% through tier 8 but rises sharply thereafter—7.8%, 40.9%, 98.0% at tiers 9–11—showing degradation is concentrated in a small population of severely-submerged clips rather than spread evenly.

Species-Count Degradation

Figure 6 shows how performance varies with the number of species present in a mixture (seed 0). Jaccard peaks at 0.914 for single-species mixtures, declines to 0.765 (2 species), 0.606 (3 species), and a minimum of 0.507 (5 species), then partially recovers at 10 species (0.660). The initial decline matches the expected difficulty of disentangling a growing number of overlapping calls; the partial recovery at 10 species is consistent with dense S4 mixtures giving the model a more redundant, information-rich context—many co-occurring species provide more corroborating acoustic evidence and let the model exploit learned priors over common high-density combinations—whereas 3–5-species mixtures fall in a harder middle regime with neither the simplicity of S1/S2 nor the redundancy of the densest S4 mixtures.

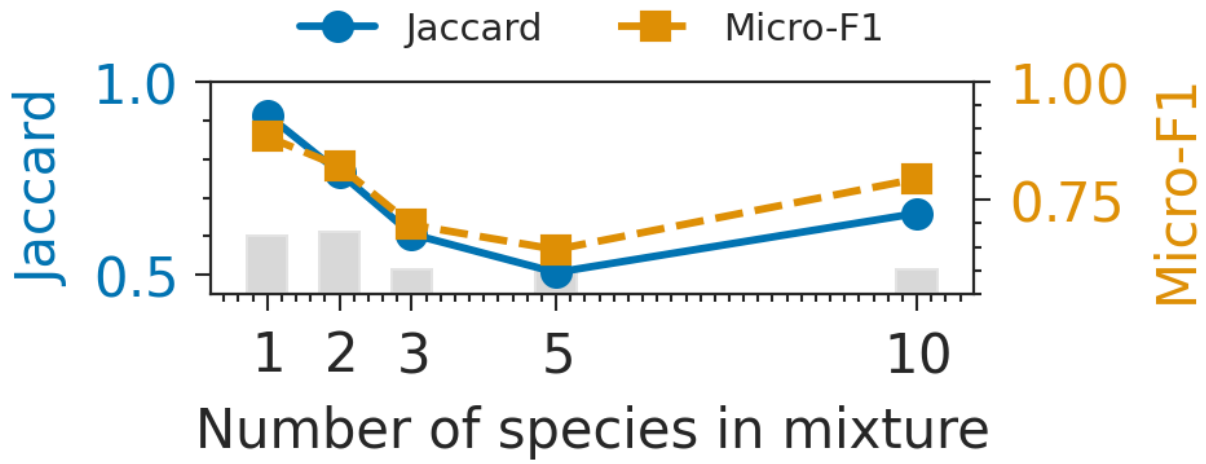
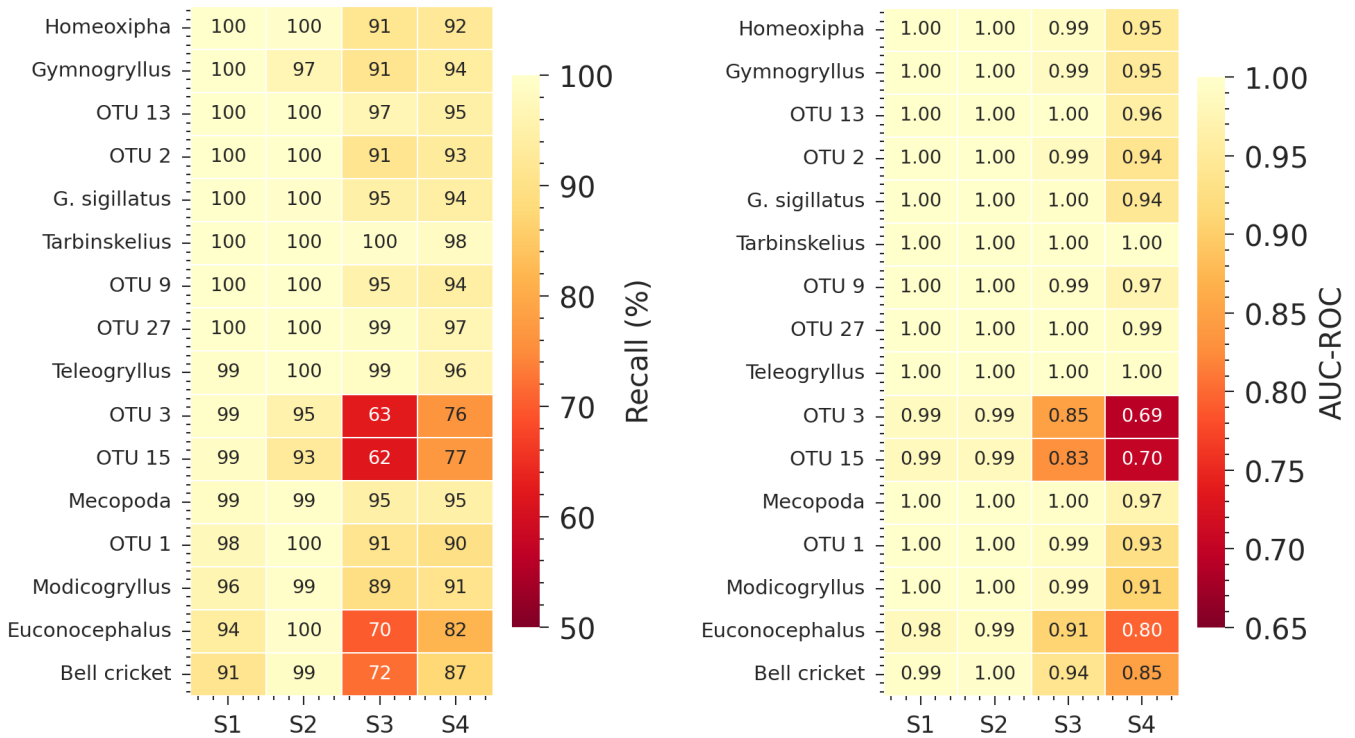


Figure 6: Jaccard and Micro-F1 as a function of the number of co-occurring species (seed 0). Grey bars indicate relative sample counts.



(a) Per-species recall (%) by scenario (seed 0). High-frequency species (*Euconocephalus*, *OTU15*, *OTU3*, *Bell cricket*, *Modicogryllus*) show lower recall in dense mixtures.

(b) Per-species AUC-ROC by scenario (3-seed probability ensemble: AUC computed on sigmoid outputs averaged across seeds), a threshold-free complement.

Figure 7: Per-species evaluation metrics comparing recall (a) and AUC-ROC (b).

Future Direction

To rigorously validate our synthetic dataset assumptions and evaluate model performance across both controlled and real-world acoustic environments, two essential empirical validation pathways are planned as upcoming future directions.

First, we will conduct physical validation within a calibrated anechoic chamber to establish precise acoustic ground truth. By eliminating ambient noise and room reflections, this controlled environment will allow us to play back reference calls through calibrated speakers at measured distances while capturing signals using the exact microphone configuration used in field deployments. Testing the model on these physical chamber recordings and comparing the outcomes to synthetic baseline predictions will allow us to assess sensor response and physical acoustic attenuation, directly informing future experimental design.

Second, to test model generalizability in unconstrained real-world scenarios, we plan to evaluate performance against expert-annotated Passive Acoustic Monitoring (PAM) recordings. We will leverage a curated set of 15-minute PAM recordings featuring high-confidence expert labels for species identification and precise call onset/offset boundaries. The model will be run directly on these recordings without fine-tuning to evaluate its out-of-box capabilities. Performance will be quantified using species-specific precision, recall, and F1-scores, alongside segment-level and event-level accuracy metrics. Crucially, a detailed error analysis will be performed to categorize false positives from ambient or non-target noise, false negatives from missed calls, and inter-species confusions. Together, these planned physical and field-based validation efforts will bridge the gap between synthetic dataset modeling and practical deployment efficacy.

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